

Acoustic UAV Detection with Pretrained Audio Embeddings: A Multi-Dataset Benchmark and an Edge-Deployable Detector

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Abstract—Passive acoustic sensing is a low-cost, unobtrusive way to detect small unmanned aerial vehicles (UAVs). Our goal is UAV classification on edge devices. To establish what an edge model should aim for, we benchmark frozen pretrained audio embeddings (AST, YAMNet, Wav2Vec 2.0, with and without triplet fine-tuning) combined with supervised and unsupervised classifiers, plus log-mel CNN and teacher–student baselines — several too heavy for edge deployment, included as reference points — on six UAV audio datasets: three public benchmarks and three of our own field recordings. Supervised linear probes on AST/YAMNet embeddings dominate (best AUC 0.759–1.000); unsupervised detectors are competitive on most corpora; and triplet fine-tuning helps mainly on our field recordings and on cross-dataset transfer. A deployable int8 mini-SE-Net (~7.4k parameters, ~22.6 KB, ~1.5–2.5 ms/sample on CPU) reaches AUC 0.68–0.99 per dataset.

Index Terms—*acoustic detection, UAV/drone detection, audio embeddings, anomaly detection, transfer learning, TinyML.*

I. INTRODUCTION

Small unmanned aerial vehicles (UAVs) are increasingly cheap and capable, raising safety, privacy, and security concerns around airports, critical infrastructure, and restricted airspace. Radar and radio-frequency systems dominate commercial counter-UAV deployments but struggle with small, autonomous, or radio-silent platforms and carry substantial cost. Passive acoustic detection is a natural complement: microphones are inexpensive and unobtrusive, and rotor noise is a strong, characteristic signal at short range.

Our aims are practical. We want to identify which models detect UAVs most accurately, to see which of them are light enough to run on edge devices, and to test all of this on our own field recordings rather than only on curated public data. The benchmark is organized around these aims: it spans representations and classifiers of very different computational cost, reports the on-device footprint and latency of the deployable model, and evaluates on three datasets we collected ourselves alongside three public ones.

Concretely, we benchmark three frozen pretrained audio embeddings — the Audio Spectrogram Transformer (AST) [1], YAMNet, and Wav2Vec 2.0 [2] — each in its off-the-shelf form and after triplet-loss fine-tuning [3], combined with a supervised logistic-regression probe and two unsupervised anomaly detectors, one-class SVM [4] and Isolation Forest [5], plus log-mel CNN and teacher–student baselines. Several configurations in this grid are too heavy for edge deployment and are included as reference points for the target on-device model. Evaluation covers six UAV audio datasets: three curated public corpora and three collected by our team in the field. Our contributions are: (i) a unified, cost-annotated benchmark of 18+ feature–classifier combinations per dataset; (ii) evidence that linear probes on AST/YAMNet embeddings dominate (AUC 0.759–1.000 across datasets) while unsupervised detectors remain competitive; and (iii) evidence that the benefit of triplet fine-tuning is dataset-dependent, concentrated on our own field recordings and on cross-dataset transfer.

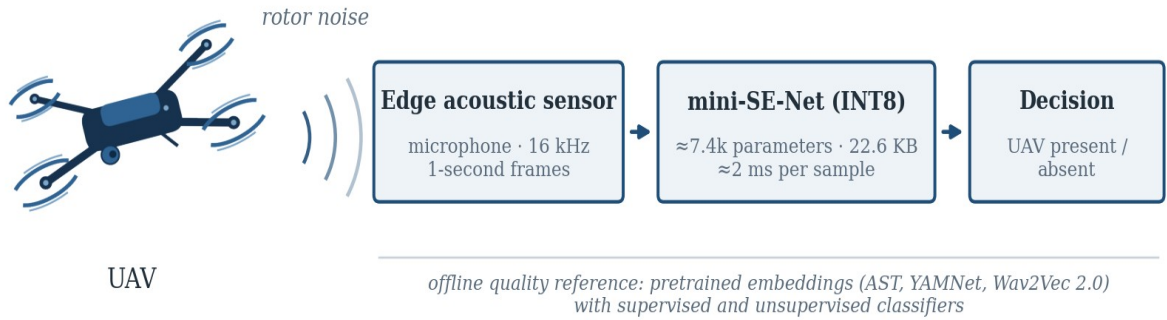


FIG. 1. System overview: rotor noise is captured by an edge acoustic sensor and classified on-device by the int8 mini-SE-Net; the embedding pipeline serves as the offline quality reference.

II. RELATED WORK

Acoustic UAV detection has progressed from classical spectral features with shallow classifiers to deep networks trained on spectrograms. In parallel, large pretrained audio models — AST [1], trained on AudioSet with a transformer architecture, YAMNet [7], a MobileNet-based AudioSet classifier, and Wav2Vec 2.0 [2], self-supervised on speech — have made high-quality general-purpose audio embeddings freely available, shifting the practical question from architecture design to representation choice. Linear probing of frozen embeddings is an established, compute-cheap way to compare such representations; triplet-loss fine-tuning [3] adapts an embedding space to a target domain using only weak same/different supervision.

On the classifier side, one-class SVM [4] and Isolation Forest [5] are standard anomaly detectors that model the normal class alone, a formulation studied extensively on industrial machine audio in the MIMII benchmark [6]. To our knowledge, no prior work benchmarks this full grid of pretrained embeddings $\times$ supervised/unsupervised classifiers across both public and field-recorded UAV corpora under one protocol.

III. DATASETS

We evaluate on six corpora (Table I). Three are public benchmarks: Binary Drone Audio [9], Multiclass Drone Audio [10], and AeroSonicDB [8]. Three were recorded by our team: Electric UAVs, FPV, and UAV, covering distinct platform families under real outdoor conditions; the field datasets are class-balanced by construction, per our data-collection protocol. Positive (UAV) examples contain rotor noise from the relevant platform family; negatives span ambient outdoor scenes and non-UAV sources. All datasets use 1-second audio clips sampled at 16 kHz. AeroSonicDB contains low-flying aircraft rather than small UAVs; we include it as a related aerial-acoustics benchmark. Each dataset is split randomly into train/validation/test partitions with no clip overlap. The corpus is documented in a formal DATASET.md datasheet. One licensing consequence is material: AeroSonicDB is distributed under CC BY-NC 4.0, so any combined corpus that includes it inherits a non-commercial restriction.

Dataset	Train (UAV/other)	Val (UAV/other)	Test (UAV/other)	Source
Binary Drone Audio	1326 (889/437)	1756 (1556/200)	1756 (1556/200)	public
AeroSonicDB	1326 (889/437)	284 (190/94)	285 (191/94)	public
Multiclass Drone Audio	1401 (701/700)	300 (150/150)	301 (150/151)	public
Electric UAVs	10648 (6112/4536)	448 (224/224)	448 (224/224)	ours
FPV	33465 (16126/17339)	4910 (1110/3800)	7334 (3534/3800)	ours
UAV	66930 (33465/33465)	14384 (7189/7195)	14365 (7176/7189)	ours

TABLE I. Datasets and split sizes. “Ours” denotes field recordings collected by the team.

IV. METHOD

A. Feature representations

Each audio clip is mapped to a fixed-dimensional vector by one of three frozen pretrained encoders: AST [1], YAMNet, or Wav2Vec 2.0 [2]. For each encoder we additionally evaluate a *triplet* variant, fine-tuned with triplet loss [3] to pull same-class clips together and push UAV/non-UAV clips apart in the embedding space. A log-mel spectrogram front end serves the trained-from-scratch baselines.

B. Classifiers

On top of each embedding we train: (i) *logistic regression*, a supervised linear probe using both classes; (ii) *one-class SVM* [4] and (iii) *Isolation Forest* [5], unsupervised detectors fit without UAV labels, scoring test clips by outlierness. Two deep baselines complete the grid: a log-mel CNN trained end-to-end, and a teacher–student autoencoder in the style of unsupervised anomaly detection on machine audio [6]. All configurations share the same splits and metrics, making the comparison a controlled study of representation and supervision level rather than of training budget. Classifier settings: logistic regression with up to 1,000 iterations; one-class SVM with $\nu = 0.05$; Isolation Forest with 100 trees and contamination 0.05; triplet fine-tuning uses 1,000 sampled triplets; the CNN and teacher–student baselines train for up to 30 epochs with batch size 32 and early stopping.

C. Metrics and cost accounting

We report accuracy, precision/recall/F1 on the UAV class, area under the ROC curve (AUC), and average precision (AP), alongside training time and per-sample CPU inference latency for every configuration. AUC and AP characterize threshold-independent ranking quality; the cost columns matter because the intended deployment is a distributed, always-on edge sensor. The quantized TinyML track completes this: a raw-waveform int8 mini-SE-Net (~7.4k parameters, ~22.6 KB) is trained and evaluated per dataset on the same test splits (Section V-E, Table IV). It ingests the raw 16,000-sample waveform directly (first convolution kernel 64, stride 8; SE reduction ratio 8) and is trained for up to 30 epochs with batch size 32 and early stopping, then post-training quantized to full int8 TFLite.

V. RESULTS

A. Best configurations per dataset

Table II reports the best configuration by AUC on each dataset. Linear probes on pretrained embeddings win everywhere: AST-based features take first place on four of six datasets, with YAMNet leading on AeroSonicDB and UAV. Performance is near-ceiling on the curated public corpora (AUC 1.000 on Multiclass, 0.999 on Binary Drone Audio) and remains strong on the harder field recordings (0.913 on Electric UAVs, 0.922 on FPV), with UAV the most difficult at 0.759. Notably, the winning configurations are also among the cheapest, with per-sample inference at 0.001–0.027 ms on CPU — the expensive part of the pipeline is the frozen encoder, not the classifier.

Dataset	Best configuration	Acc	F1†	AUC	ms/samp
Binary Drone Audio	AST · LogReg	0.996	0.998	0.999	0.026
AeroSonicDB	YAMNet · LogReg	0.923	0.943	0.954	0.017
Multiclass Drone Audio	AST · LogReg	0.997	0.997	1.000	0.004
Electric UAVs	AST-triplet · LogReg	0.792	0.817	0.913	0.002
FPV	AST-triplet · LogReg	0.850	0.846	0.922	0.003
UAV	YAMNet-triplet · LogReg	0.702	0.712	0.759	0.001
Train-all → FPV	AST-triplet · LogReg	0.998	0.998	1.000	0.002

TABLE II. Best configuration by AUC per dataset. †F1 on the UAV class.

B. Supervised vs. unsupervised detection

Table III compares the best supervised probe against the best unsupervised detector on each dataset. The gap is small on most corpora (0.999 vs. 0.948 on Binary, 1.000 vs. 0.973 on Multiclass, 0.954 vs. 0.864 on AeroSonicDB) and effectively zero on the field recordings: Electric UAVs (0.913 vs. 0.904), FPV (0.922 vs. 0.911), and UAV (0.759 vs. 0.747). The exception pattern is informative too: unsupervised detectors on *raw* (non-triplet) Wav2Vec 2.0 features

collapse on several datasets, indicating that the speech-pretrained embedding is poorly aligned with rotor acoustics unless adapted.

Dataset	Best supervised	AUC	Best unsupervised	AUC
Binary Drone Audio	AST · LogReg	0.999	AST · IsoForest	0.948
AeroSonicDB	YAMNet · LogReg	0.954	AST-triplet · IsoForest	0.864
Multiclass Drone Audio	AST · LogReg	1.000	AST · IsoForest	0.973
Electric UAVs	AST-triplet · LogReg	0.913	AST-triplet · OCSVM	0.904
FPV	AST-triplet · LogReg	0.922	AST-triplet · IsoForest	0.911
UAV	YAMNet-triplet · LogReg	0.759	YAMNet-triplet · OCSVM	0.747
Train-all → FPV	AST-triplet · LogReg	1.000	AST-triplet · OCSVM	0.992

TABLE III. *Best supervised vs. best unsupervised configuration per dataset (AUC).*

C. Effect of triplet fine-tuning

To test whether triplet adaptation generalizes, we compare each frozen encoder–classifier pair with its triplet counterpart on every dataset. The effect is dataset-dependent rather than universal. On Electric UAVs and UAV, triplet variants improve 8 of 9 combinations each (mean AUC gain of +0.15 and +0.17), and in the pooled cross-dataset run they improve all 9 combinations. On FPV the picture is split: AST-triplet sharply lifts the unsupervised detectors (Isolation Forest 0.483 → 0.911, OCSVM 0.554 → 0.895) and ties the full AST probe (0.922 vs. 0.921), while the YAMNet and Wav2Vec 2.0 triplet variants underperform their frozen counterparts. On the public corpora adaptation is mostly harmful — it degrades 7 of 9 combinations on Binary Drone Audio and all 9 combinations on Multiclass Drone Audio — with AeroSonicDB the exception, where it improves 5 of 6 unsupervised combinations while weakening the supervised probes. The most consistent positive effect is therefore on unsupervised detectors over our own recordings, where adaptation turns near-chance OCSVM and Isolation Forest scores into competitive ones.

D. Dataset difficulty

Ranking datasets by best-achievable AUC exposes a clear difficulty ordering: Multiclass (1.000) ≈ Binary Drone Audio (0.999) ≫ AeroSonicDB (0.954) > FPV (0.922) > Electric UAVs (0.913) > UAV (0.759). The three hardest corpora are precisely our field recordings, despite being the largest (10.6k–33.5k training clips) — evidence that acoustic diversity and recording conditions, not data volume, are the binding constraint. Cross-dataset generalization (train on the five other corpora, test on the FPV test split) is strong: AST-triplet · LogReg reaches AUC 1.000, and even the frozen AST probe (0.996) and the log-mel CNN (0.997) transfer well — evidence that drone acoustics learned on other platforms carry over to FPV. Unsupervised detectors on frozen embeddings collapse in this setting (AUC 0.37–0.71) unless triplet-adapted (0.95–0.99), so the adaptation is what makes label-free detection viable across corpora.

E. On-device model

Table IV and Fig. 2 report the int8 mini-SE-Net (raw waveform in, ~7.4k parameters) per dataset. The model fits in ~22.6 KB everywhere and runs at 1.5–2.5 ms per sample. Detection quality tracks the difficulty ordering of Section V-D: near the embedding ceiling on the curated public corpora (0.989 on Binary Drone Audio, 0.975 on Multiclass, within 0.01–0.13 AUC of the best embedding configuration), but with a wider gap on the field recordings (0.861 on FPV, 0.701 on UAV, 0.683 on Electric UAVs) and a marked drop on the pooled cross-dataset run (0.583). Two caveats: the Electric UAVs run used augmented training data, unlike the other runs, and all latencies were measured with the TFLite interpreter on a Colab host CPU rather than on the target device.

Dataset	Acc	F1†	AUC	Size (KB)	ms/samp
Binary Drone Audio	0.950	0.951	0.989	22.6	2.299
Multiclass Drone Audio	0.900	0.899	0.975	22.6	2.219
FPV	0.774	0.780	0.861	22.6	1.899
AeroSonicDB	0.803	0.781	0.826	22.6	2.017

Dataset	Acc	F1 $\uparrow$	AUC	Size (KB)	ms/samp
UAV	0.651	0.680	0.701	22.6	1.454
Electric UAVs (aug.)	0.679	0.593	0.683	22.9	1.865
Train-all $\rightarrow$ FPV	0.525	0.302	0.583	22.9	2.518

TABLE IV. *Int8 mini-SE-Net per dataset. $\uparrow$ F1 on the UAV class. Latency measured on a Colab host CPU.*

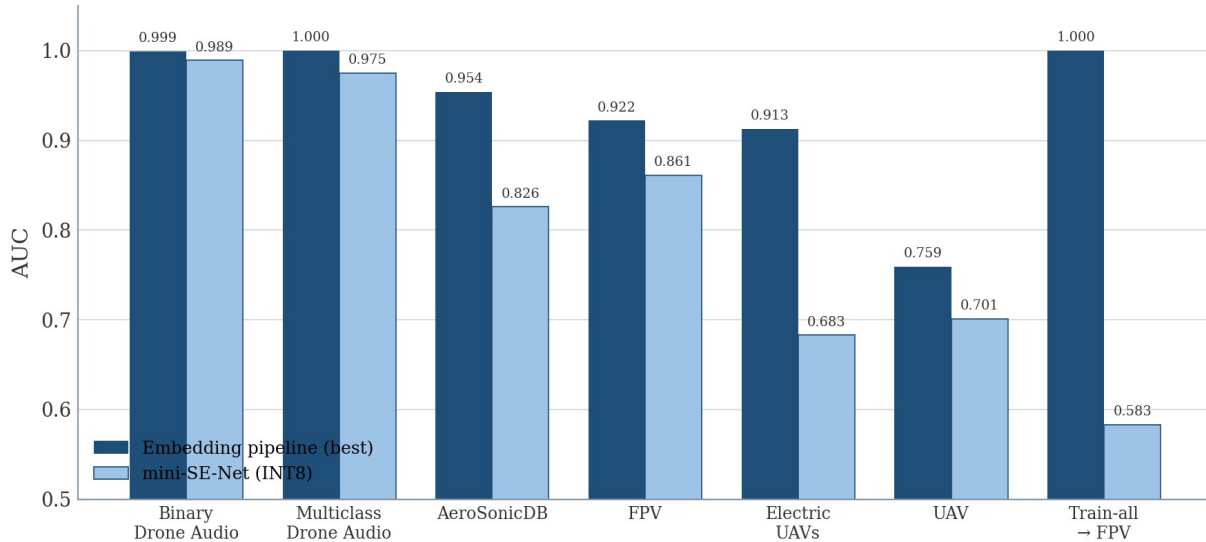


FIG. 2. *Best embedding-pipeline AUC vs. int8 mini-SE-Net AUC per dataset.*

VI. DISCUSSION AND LIMITATIONS

Two limitations frame these results. First, the on-device latency of the int8 mini-SE-Net (~ 1.5 – 2.5 ms/sample) was measured with the TFLite interpreter on a Colab host CPU, not yet on the target embedded device, so real deployment timing remains to be confirmed; its accuracy also trails the embedding ceiling by a wide margin on the hardest corpora (e.g., 0.68 on Electric UAVs vs. 0.913 for the best embedding), leaving clear room to improve. Second, the non-commercial license inherited from AeroSonicDB restricts commercial use of any combined corpus containing it.

Two observations stand on the completed evidence: AST-family features are the strongest representation on four of six datasets, and unsupervised detectors paired with strong embeddings remain competitive with supervised probes on most corpora.

VII. CONCLUSION AND FUTURE WORK

We presented a unified benchmark of pretrained audio embeddings with supervised and unsupervised classifiers for acoustic UAV detection across six datasets. Frozen AST and YAMNet embeddings with linear probes achieve AUC 0.759–1.000; unsupervised detectors on the same embeddings remain competitive; and triplet fine-tuning helps most consistently on our own field recordings and on cross-dataset transfer, with mixed effects on the public corpora. A deployable int8 mini-SE-Net reaches AUC 0.68–0.99 across the single datasets while fitting in ~ 22.6 KB. Future work benchmarks this on-device model on the target embedded hardware and narrows its accuracy gap to the embedding ceiling on the hardest corpora.

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